

Dated:

DBS 1 Week #9

24K0756

1. Entities:

- Customer
- Booking
- Vehicle
- Branch
- Employee
- Payment
- Maintenance
- Service
- ↳ Additional

2. Entities & their Attributes:

- Customer — customer-id (PK), name, email, phone, license-no., address, DOB
- Booking — booking-id (PK), start-date, end-date, status, total-amount, customer-id (FK), vehicle-id (FK), employee-id (FK)
- Vehicle — vehicle-id (PK), license-plate, make, model, year, type, daily-rate, status, Branch-id
- Branch — branch-id (PK), branch-name, city, address, phone, manager-name
- Employee — employee-id (PK), name, role, email, phone, branch-id (FK)
- Payment — payment-id (PK), amount, pay-date, pay-method, status, Booking-id (FK)
- Maintenance — maintenance-id (PK), service-date, service-type, description, cost, technician, vehicle-id
- Additional Service — service-id (PK), service-name, description, daily-cost

Dated:

24K0756

3. Entity Relationships & Cardinalities:

Relationship	Type	
Booking → Customer	1:M	One customer, many bookings everytime.
Booking → Vehicle	M:1	Many bookings can be for 1 vehicle.
Vehicle → Branch	M:1	A branch manages multiple vehicle.
Employee → Branch	M:1	Multiple employees / branch
Employee → Booking	1:M	1 emp. handles many bookings.
Booking → Payment	1:1	Each booking has exactly 1 payment.
Vehicle → Maintenance	1:M	1 vehicle has many maintenance records.
Booking ↔ Additional Service	M:N	A booking can have multiple services & viceversa.

